

Assignment 3 Report: End-to-End Hugging Face Model Training & Docker Deployment

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Course: DL Ops

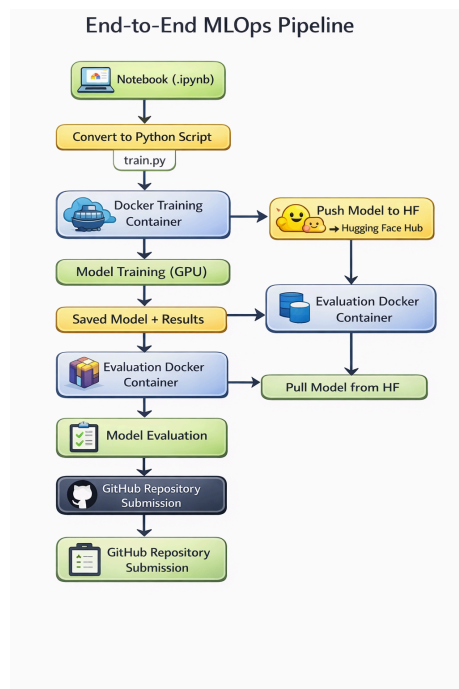
Institute: IIT Jodhpur

1. Objective

The goal of this assignment was to build and execute a full end-to-end Machine Learning Operations (MLOps) pipeline. This workflow covers model development and validation, containerization with Docker, deployment through the Hugging Face Hub, and experiment tracking with GitHub to maintain reproducibility and ease of deployment across environments.

2. Overall Workflow

The complete workflow followed in this assignment is outlined below in the figure:



3. Environment Setup Using Docker

Docker was used to create an isolated and reproducible environment independent of the host system.

Steps Performed:

- Created Dockerfile
- Installed dependencies
- Configured working directory
- Enabled GPU support

Docker Build Command:

```
docker build -t hf-train:v1 .
```

Docker Run Command:

```
docker run -it --rm \
--gpus all \
--shm-size=8g \
-v $(pwd):/workspace \
hf-train:v1
```

4. Notebook Conversion to Python Script

The instructor-provided notebook was converted into a production-ready Python script.

Steps:

1. Downloaded notebook
2. Converted into Python script
3. Removed notebook artifacts
4. Organized execution flow

Example command used:

```
Bash
jupyter nbconvert --to python notebook.ipynb
```

Final script used: train.py

5. Model Selection

A pre-trained transformer model (**DistilBERT**) from Hugging Face was selected.

Reasons for Selection:

- Lightweight architecture
- Faster training compared to standard BERT
- Good performance for text classification tasks
- Efficient fine-tuning capability

Note: The tokenizer and model were loaded directly from Hugging Face.

6. Model Training Using Hugging Face Trainer API

The model was fine-tuned using the Hugging Face Trainer API.

Training Configuration:

Parameter	Value
Epochs	3
Batch Size (Training)	10
Batch Size (Evaluation)	16
Learning Rate	2.00E-05
Optimizer	AdamW
Evaluation Strategy	Epoch-wise Evaluation
Device	GPU (CUDA enabled)

Training included:

- Dataset preprocessing
- Tokenization
- Trainer configuration
- GPU-accelerated training

7. Model Evaluation

After training, the model was evaluated on the validation/test dataset.

Metrics Used:

- Accuracy
- F1 Score
- Loss

	precision	recall	f1-score	support
children	0.61	0.60	0.60	200
comics_graphic	0.75	0.72	0.73	200
fantasy_paranormal	0.31	0.23	0.27	200
history_biography	0.51	0.52	0.52	200
mystery_thriller_crime	0.52	0.45	0.48	200
poetry	0.67	0.81	0.73	200
romance	0.55	0.58	0.56	200
young_adult	0.35	0.40	0.37	200
accuracy			0.54	1600
macro avg	0.53	0.54	0.53	1600
weighted avg	0.53	0.54	0.53	1600

Example Output:

```

accuracy          0.53    0.54    0.53    1000
macro avg         0.53    0.54    0.53    1000
weighted avg      0.53    0.54    0.53    1000

tokenizer_config.json: 100%
vocab.txt: 100%
tokenizer.json: 100%
config.json: 100%
model.safetensors: 100%
Loading weights: 100%
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-cased
Key | Status | Details
-----
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.bias | UNEXPECTED |
vocab_transform.weight | UNEXPECTED |
vocab_projector.bias | UNEXPECTED |
vocab_layer_norm.bias | UNEXPECTED |
pre_classifier.bias | MISSING |
classifier.weight | MISSING |
classifier.bias | MISSING |
pre_classifier.weight | MISSING |

Notes:
- UNEXPECTED : can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING : those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

```

The results confirmed the successful fine-tuning of the pre-trained model.

8. Saving and Uploading Model to Hugging Face

A Hugging Face account was created, and an access token was generated to push the model to the Hub.

Login Command:

```
python
from huggingface_hub import login
login()
```

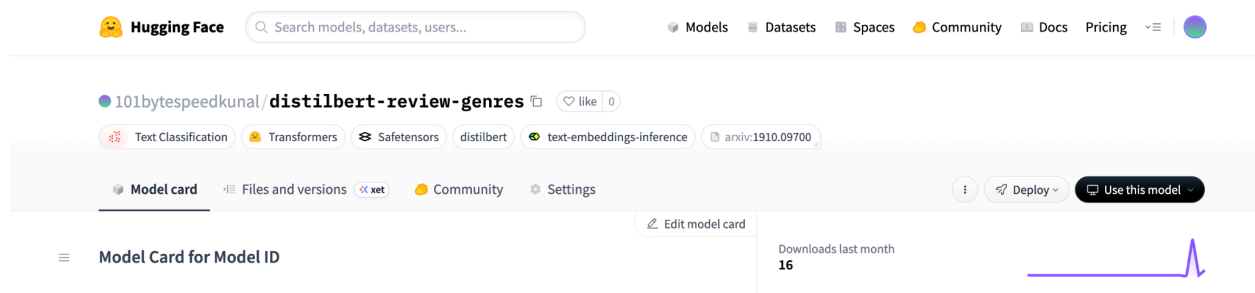
Model Upload:

```
python
model.push_to_hub("101bytespeedkunal/distilbert-review-genres")
```

The following artifacts were uploaded:

- Model weights
- Tokenizer
- Configuration files

Hugging Face Model Link: [Link](#)



9. Re-evaluation from Hugging Face Repository

A separate evaluation Docker container was created, which automatically downloaded the model from Hugging Face and executed the evaluation.

Evaluation Image Build:

```
Bash
docker build -t hf-eval:v1 -f Dockerfile.eval .
```

Run Evaluation:

```
Bash
docker run -it --rm --gpus all hf-eval:v1
```

Output:

Plaintext

Model loaded successfully from Hugging Face

Observation: The evaluation results were consistent with local training results, confirming correct deployment.

10. Final Evaluation Docker Image

A lightweight production Docker image was created specifically for inference purposes.

Purpose:

- Isolate training and inference environments
- Set up a consistent and reproducible evaluation pipeline
- Prepare the model for production-grade deployment

11. GitHub Repository

All project files were version-controlled and pushed to GitHub.

Repository Includes:

- train.py
- eval.py
- Dockerfile
- Dockerfile.eval
- requirements.txt
- README.md

✓ GitHub Repository Link: [Link](#)

The screenshot shows a GitHub repository page for the user 'Givhel' and repository 'MLOps_KunalMishra_M25CSA036'. The repository is public and has 5 branches and 0 tags. The current branch is 'Assignment_2', which is 1 commit ahead of and 1 commit behind the 'main' branch. The repository contains 9 files, all submitted as 'Assignment 2 submission' 5 days ago. The files are: .gitignore, Dockerfile, Dockerfile.eval, README.md, eval.py, genre_reviews_dict.pickle, requirements.txt, train.py, and upload_model.py. The repository has 0 stars, 0 forks, and 0 releases. The 'About' section states 'No description, website, or topics provided.' The 'Releases' section states 'No releases published' with a link to 'Create a new release'. The 'Packages' section states 'No packages published' with a link to 'Publish your first package'. The 'Deployments' section shows 7 deployments, with 'github-pages' deployed last month.

File	Commit	Time
.gitignore	Assignment 2 submission	5 days ago
Dockerfile	Assignment 2 submission	5 days ago
Dockerfile.eval	Assignment 2 submission	5 days ago
README.md	Assignment 2 submission	5 days ago
eval.py	Assignment 2 submission	5 days ago
genre_reviews_dict.pickle	Assignment 2 submission	5 days ago
requirements.txt	Assignment 2 submission	5 days ago
train.py	Assignment 2 submission	5 days ago
upload_model.py	Assignment 2 submission	5 days ago

12. Challenges Faced

During implementation, several challenges were encountered:

- Package incompatibilities within Docker containers
- GPU setup and configuration problems
- Absent or unresolved dependencies during container runtime
- Docker image size optimization and control

These issues were resolved through iterative debugging and environment configuration.

13. Key Learnings

This assignment provided practical exposure to:

- Complete MLOps pipeline implementation
- Model packaging and isolation with Docker
- Deploying models via Hugging Face Hub
- Ensuring reproducibility in machine learning experiments
- Source code and experiment tracking with GitHub
- Decoupling training and inference environments

14. Conclusion

The assignment effectively showcased the full machine learning lifecycle, from initial experimentation to final deployment. Docker was used to maintain reproducibility, Hugging Face facilitated seamless model sharing, and GitHub ensured organized version control, together establishing a robust and production-ready MLOps pipeline.